

Ashly Agrawal

Indore, Madhya Pradesh

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EDUCATION

Vellore Institute of Technology University

B.Tech in Computer Science and Engineering

- Relevant Coursework: Applied Mathematics, Database Systems, Data Structures & Algorithms, Machine Learning, Cloud Computing, Software Engineering

Sehore, Madhya Pradesh

Aug 2023 – May 2027

EXPERIENCE

Software Engineering Intern

Mahamai Hydro Power Pvt. Ltd.

Mar. 2025 – May 2025

Gurugram, Haryana

- Designed and authored 20+ end-to-end data pipeline specifications across 4 services, defining schemas and contracts that eliminated drift between producer and consumer systems.
- Built regression-based forecasting models in Python and profiled production databases to isolate throughput bottlenecks, then implemented schema-level fixes that doubled query speed.
- Developed internal reporting and monitoring scripts to surface KPI baselines and recurring operational failures, cutting resolution time by 40% through automated triage.
- Collaborated with engineering teams to translate business logic into database schema changes and RESTful API consumers, strengthening data integrity across 5 facilities.

SKILLS

Languages: Python, C++, Java, JavaScript/TypeScript, SQL, HTML/CSS

Frontend: React, Next.js, Tailwind CSS, Redux/SWR, Formik

Backend: Node.js, Express, REST APIs, JWT/Passport Authentication, Server Actions

Databases & Cloud: MongoDB, PostgreSQL (Supabase), MySQL, AWS (EC2, IAM), Docker

Engineering Practices: Git, CI basics, Data Pipeline Design, Statistical Modeling, EDA

PROJECTS

Dressly – Full-Stack E-Commerce Platform | Next.js, Node.js, Express, MongoDB – GitHub | Live Demo 2025

- Built a full-stack e-commerce application with a Next.js frontend and a modular Express/MongoDB backend, separating routes, controllers, and Mongoose models for maintainability.
- Implemented JWT-based authentication with Passport.js, including protected-route middleware and role-based authorization for customer and admin access.
- Integrated Stripe and PayPal SDKs to handle end-to-end checkout, and built cart, wishlist, and order-management features through a dedicated API service layer using Axios and SWR.

Acadence – AI Learning Companion SaaS | Next.js, Supabase, Clerk, Vapi AI – GitHub | Live Demo 2025

- Developed a real-time AI tutoring platform using Next.js App Router and Server Actions, removing the need for a separate REST layer for core database operations.
- Integrated Vapi's real-time voice AI SDK to power live conversational tutoring sessions, managing multi-state call logic (connecting, active, speaking) and persisting session history.
- Implemented authentication and tiered subscription billing with Clerk, including plan-based feature gating, and designed a Supabase (PostgreSQL) schema for companions, bookmarks, and sessions.

Brain Tumor Detection | Lead Researcher | Python, scikit-learn, NumPy – GitHub 2025

- Directed a 2-person team through a full analytical pipeline on 3,000+ records, from data ingestion to validation and evaluation.
- Applied normalization and stratified data splits with rigorous quality controls, achieving 90%+ held-out model accuracy.

CERTIFICATIONS

AWS Certified Cloud Practitioner, Amazon Web Services, 2025

Oracle Certified Java Developer, Oracle, 2025

Introduction to IoT, NPTEL, 2023

ACHIEVEMENTS

Goldman Sachs CS India Hackathon: Ranked 448 out of 40,000+ participants nationwide, 2025

ISRO Experiential Learning Program: Selected for satellite imagery training at ISRO-NRSC Hyderabad; one of a limited cohort chosen nationwide, Oct. 2025

GirlScript Summer of Code: Mentored contributors and reviewed 8+ pull requests across 2 repositories; enforced quality standards and kept branches in sync, Aug. 2024

Pupil, Codeforces: Ranked top 1,500 of 30,000+ globally in competitive programming, Feb. 2026